

# ALTROS

Autonomous Learning & Tactical Reasoning Operating System

## Complete Build Guide — From Zero to Advanced AI Assistant

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| Feature                  | Status      |
|--------------------------|-------------|
| Local AI Brain (Llama3)  | Implemented |
| 3-Layer Memory System    | Implemented |
| Voice Mode + Wake Word   | Implemented |
| Internet Search Pipeline | Implemented |
| PC Control (35+ Apps)    | Implemented |
| Phone Access (Jarvis UI) | Implemented |
| WhatsApp Integration     | Implemented |
| Gmail Integration        | Implemented |
| Instagram Integration    | Implemented |
| Auto-Startup             | Implemented |

**Total Cost: Rs. 0 | Privacy: 100% Local | Platform: Windows 11**

# Chapter 1: What is ALTROS?

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ALTROS (Autonomous Learning and Tactical Reasoning Operating System) is a fully local, private, and free personal AI assistant built from scratch. It runs entirely on your laptop using open-source models, with no cloud dependency and no subscription fees.

ALTROS is not a chatbot. It is a complete AI system that can control your PC, access the internet in real-time, remember your life, read books, send WhatsApp messages, manage Gmail, control Instagram, and have voice conversations — all while keeping your data 100% private on your own machine.

## Core Capabilities

- Natural Hinglish conversation with personality
- Long-term memory — remembers your goals, habits, preferences
- Book learning via RAG (Retrieval Augmented Generation)
- Real-time internet search with smart filtering
- PC control — open apps, create folders, take screenshots
- Voice I/O with interrupt detection (say something = ALTROS stops)
- Wake word activation — say 'computer' to activate
- Phone access via browser — Jarvis-style mobile UI
- WhatsApp — send/receive messages and files
- Gmail — read, send, delete, search emails
- Instagram — post, DM, like, follow, search profiles
- Autonomy — daily briefing, tasks, reminders

# Chapter 2: Hardware Requirements

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## Recommended Hardware (Used in This Build)

| Component | Spec                | Purpose                  |
|-----------|---------------------|--------------------------|
| CPU       | Intel i7 13th Gen   | Running Python + Whisper |
| GPU       | NVIDIA RTX 4050 6GB | Accelerating Llama3      |
| RAM       | 16GB DDR5           | Llama3 needs ~8GB RAM    |
| Storage   | 512GB SSD           | Model files + project    |
| OS        | Windows 11          | Primary platform         |
| Internet  | Any broadband       | Web search + ngrok       |

## Minimum Hardware

- CPU: Any modern i5/Ryzen 5 (6+ cores)
- RAM: 8GB minimum (16GB recommended)
- GPU: Not required but speeds up Llama3 significantly
- Storage: 20GB free space (Llama3 = 4.7GB)
- OS: Windows 10/11 (Linux also works with minor changes)

## Chapter 3: Software Installation — Step by Step

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### Step 1: Install Python 3.11+

Download from: <https://python.org/downloads>

- Download Python 3.11 or higher
- During installation: CHECK 'Add Python to PATH'
- Verify installation:

```
python --version  
pip --version
```

### Step 2: Install Visual Studio Code

Download from: <https://code.visualstudio.com>

- Install Python extension inside VS Code
- This will be your main code editor

### Step 3: Install Ollama (Local AI Engine)

Download from: <https://ollama.com>

- Install Ollama for Windows
- After installation, open terminal and run:

```
ollama pull llama3
```

- This downloads Llama3 8B model (~4.7GB) — takes 10-20 minutes
- Start Ollama server:

```
ollama serve
```

**NOTE:** Keep 'ollama serve' running in a separate terminal whenever using ALTROS.

### Step 4: Install Node.js (for WhatsApp)

Download from: <https://nodejs.org>

- Download LTS version (.msi installer, NOT zip)
- Install with default settings
- Verify:

```
node --version  
npm --version
```

## Step 5: Install Ngrok (for Phone Access)

Download from: <https://ngrok.com/download>

- Create free account at [ngrok.com](https://ngrok.com)
- Download ngrok for Windows
- Add authtoken:

```
ngrok config add-authtoken YOUR_TOKEN_HERE
```

**NOTE:** Authtoken milega: <https://dashboard.ngrok.com/auth/tokens>

## Step 6: Install All Python Libraries

Open terminal and run these commands one by one:

```
pip install requests
pip install chromadb
pip install pypdf
pip install openai-whisper
pip install sounddevice
pip install numpy
pip install pytsx3
pip install pvporcupine
pip install keyboard
pip install fastapi
pip install uvicorn
pip install ddgs
pip install pyautogui
pip install pillow
pip install google-auth-oauthlib
pip install google-auth-httplib2
pip install google-api-python-client
pip install instagrapi
pip install aiofiles
```

## Step 7: Install WhatsApp Node.js Packages

```
mkdir C:\ALTROS\whatsapp
cd C:\ALTROS\whatsapp
npm init -y
npm install whatsapp-web.js qrcode-terminal
```

## Chapter 4: Project Folder Structure

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Create this exact folder structure at C:\ALTROS\

```
C:\ALTROS\  
|-- main.py          (Entry point)  
|-- config.py        (All settings)  
|-- credentials.json  (Google OAuth - you download this)  
|-- requirements.txt  
|-- start_altros.bat  (Auto-start script)  
|  
|-- core/  
|   |-- __init__.py  
|   |-- brain.py      (LLM engine)  
|   |-- memory.py     (3-layer memory)  
|   |-- router.py     (Module router)  
|   |-- thinking.py   (Intent + planning)  
|  
|-- modules/  
|   |-- __init__.py  
|   |-- base.py       (Base module class)  
|   |-- chat.py       (Fallback chat)  
|   |-- knowledge.py  (RAG system)  
|   |-- voice.py      (Whisper + pyttsx3)  
|   |-- wake_word.py  (Porcupine)  
|   |-- internet.py   (DDGS search)  
|   |-- pc_control.py (App + file control)  
|   |-- server.py     (FastAPI server)  
|   |-- autonomy.py   (Tasks + reminders)  
|   |-- instagram.py  (Instagram API)  
|   |-- gmail.py      (Gmail API)  
|   |-- whatsapp.py   (WhatsApp bridge)  
|  
|-- whatsapp/  
|   |-- whatsapp_bridge.js  
|   |-- package.json  
|   |-- node_modules/  
|  
|-- static/  
|   |-- index.html    (Jarvis UI)  
|  
|-- data/  
|   |-- memory/  
|       |-- user_profile.json  
|       |-- facts.json  
|       |-- tasks.json  
|       |-- gmail_token.json  
|       |-- instagram_session.json  
|       |-- conversations/
```

```
| -- knowledge/  
    |-- chromadb/
```

## Chapter 5: Configuration (config.py)

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This is the most important file. All settings are here. Edit only this file for customization.

### Key Settings

```
USER_NAME      = 'YourName'
ASSISTANT_NAME = 'ALTROS'
OLLAMA_MODEL   = 'llama3'
OLLAMA_BASE_URL = 'http://localhost:11434'
WAKE_WORD      = 'computer'
PORCUPINE_ACCESS_KEY = 'your_key_here'
```

### Module Toggles — All Must Be True

```
MODULES_ENABLED = {
    'chat'      : True,
    'knowledge' : True,
    'autonomy'  : True,
    'voice'     : True,
    'wake_word' : True,
    'internet'  : True,
    'pc_control': True,
    'server'    : True,
    'instagram' : True,
    'gmail'     : True,
}
```

**NOTE:** Agar koi module 'False' hai toh wo kaam nahi karega. Sab True rakho.

### Porcupine Wake Word Key

Free key milegi: <https://console.picovoice.ai>

- Account banao -> AccessKey copy karo -> config.py mein paste karo

# Chapter 6: How to Run ALTROS

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## Manual Start (3 Terminals)

### Terminal 1 — Ollama

```
ollama serve
```

### Terminal 2 — ALTROS

```
cd C:\ALTROS  
python main.py
```

### Terminal 3 — Ngrok (Phone Access)

```
ngrok http 8000
```

### Terminal 4 — WhatsApp (Optional)

```
cd C:\ALTROS\whatsapp  
node whatsapp_bridge.js
```

## Auto-Start Setup (Recommended)

Create start\_altros.bat in C:\ALTROS\:

```
@echo off  
start '' 'ollama' serve  
timeout /t 3  
start '' cmd /k 'cd /d C:\ALTROS && python main.py'  
timeout /t 5  
start '' cmd /k 'ngrok http 8000'
```

Add to Windows Startup: Win+R -> shell:startup -> paste shortcut

## Phone Access

- After ngrok starts, open <http://localhost:4040> in browser
- Copy the ngrok URL (example: <https://abc123.ngrok-free.app>)
- Open that URL on your phone — Jarvis UI will load
- Tap 'Voice Mode' for hands-free conversation

## Chapter 7: Complete Commands Reference

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### System Commands (in terminal)

| Command              | Action                   |
|----------------------|--------------------------|
| exit                 | ALTROS band karo         |
| clear                | Session clear karo       |
| memory               | Profile dekho            |
| books                | Knowledge base count     |
| tasks                | Pending tasks dekho      |
| voice                | Mic se baat karo (5 sec) |
| voice 10             | 10 sec record karo       |
| help                 | Commands list dekho      |
| remember naam Manish | Kuch yaad karwao         |

### PC Control Commands (say or type)

- chrome kholo / open chrome
- vs code kholo
- screenshot lo
- volume up / volume down / mute
- shutdown / restart
- desktop pe 'ProjectName' folder banao
- notepad kholo
- calculator kholo
- task manager kholo

### Internet Search Commands

- latest news batao
- iPhone 16 price India mein kya hai
- aaj ka weather Mathura mein
- Bitcoin price abhi kya hai
- poco x7 pro specs batao

### Memory Commands

- remember goal software engineer banna
- remember hobby volleyball
- remember project ALTROS bana raha hoon
- mera profile dikho

## Gmail Commands

- inbox dekho
- unread mail dikho
- latest mail padho
- mail bhejo to xyz@gmail.com subject "Hello" body "Kya haal hai?"
- gmail search invoice
- latest mail delete karo
- reply "Theek hai, kal milte hain"

## Instagram Commands

- instagram login username password
- post karo C:\photo.jpg caption "Meri photo"
- dm kar @username "Hello yaar!"
- inbox dekho instagram
- search kar @virat.kohli
- mera profile dikho
- follow karo @username
- unfollow @username
- latest post delete karo

## WhatsApp Commands

After QR scan / phone number link:

- WhatsApp pe: 'altros latest news batao' — ALTROS reply karega
- Python se: number pe message bhejo

## Chapter 8: Module Details

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### **brain.py — LLM Engine**

Ollama API se Llama3 model ko call karta hai. System prompt, memory context, aur internet search results ko combine karke response generate karta hai.

### **memory.py — 3-Layer Memory**

Layer 1: Short-term (last 20 messages). Layer 2: Long-term facts with importance score (1-10). Layer 3: ChromaDB semantic memory for books. JSON files mein persist hota hai.

### **router.py — Module Router**

User input ko sahi module pe route karta hai. PC Control aur Autonomy ke liye brain bypass hota hai — direct response aata hai.

### **thinking.py — Thinking Engine**

Intent detect karta hai (question/task/command), plan banata hai complex queries ke liye, self-check karta hai.

### **internet.py — Internet Search**

DDGS (DuckDuckGo) se search karta hai. Query optimize karta hai, results filter karta hai, duplicates remove karta hai.

### **pc\_control.py — PC Control**

35+ apps open kar sakta hai. Folders/files create, delete, copy, move, rename. Screenshot, volume, shutdown. Security layer se dangerous commands block.

### **voice.py — Voice I/O**

Whisper small model se speech-to-text. pyttsx3 se text-to-speech. Offline — no cloud.

### **wake\_word.py — Wake Word**

Pvporcupine se 'computer' word detect karta hai background mein. Callback trigger hota hai.

### **server.py — Web Server**

FastAPI + Uvicorn on port 8000. /chat endpoint. /chat/stream SSE streaming. /status, /memory endpoints.

### **autonomy.py — Autonomy**

Daily briefing on startup. Task management with priority scores. Reminder system with threading.

### **gmail.py — Gmail**

Google Gmail API v1. OAuth2 authentication. Read, send, delete, search, reply.

### **instagram.py — Instagram**

instagrapi library. Post photos/videos, send DMS, like, follow, unfollow, search profiles.

### **whatsapp.py — WhatsApp**

Python side of bridge. Communicates with Node.js bridge on port 8001.

# Chapter 9: Third-Party Integrations Setup

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## 9.1 Gmail Setup

Step 1: Google Cloud Console

- Go to: <https://console.cloud.google.com>
- Create New Project -> Name: ALTROS-Gmail
- APIs and Services -> Enable APIs -> Search 'Gmail API' -> Enable

Step 2: Create Credentials

- APIs and Services -> Credentials -> Create Credentials -> OAuth 2.0 Client ID
- Application type: Desktop App
- Download credentials.json
- Place credentials.json in C:\ALTROS\

Step 3: First Login

- Run `python main.py`
- Browser will open automatically for Google login
- Login and allow permissions
- `gmail_token.json` will be saved automatically

**NOTE:** After first login, token auto-refreshes. No manual login needed again.

## 9.2 Instagram Setup

- Run ALTROS: `python main.py`
- Type: `instagram login yourusername yourpassword`
- Session saved to `instagram_session.json`
- No re-login needed after this

**NOTE:** Use your real Instagram account. Avoid too many actions per day to prevent ban.

## 9.3 WhatsApp Setup

- Terminal: `cd C:\ALTROS\whatsapp`
- Run: `node whatsapp_bridge.js`
- QR code appears in terminal
- WhatsApp phone -> Settings -> Linked Devices -> Link a Device -> Scan QR
- OR: Use 'Link with phone number' if camera is broken

**NOTE:** WhatsApp session saved in `wa_session` folder. QR scan only needed once.

## Chapter 10: Teaching ALTROS Books

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ALTROS can read any PDF and answer questions from it. This uses RAG (Retrieval Augmented Generation) with ChromaDB vector database.

### How to Add Books

```
python main.py --add-book "C:\path\to\book.pdf"
```

### Example

```
python main.py --add-book "C:\Users\manish\Downloads\Atomic Habits.pdf"  
python main.py --add-book "C:\Users\manish\Downloads\Psychology of Persuasion.pdf"  
python main.py --add-book "C:\Users\manish\Downloads\Thinking Fast and Slow.pdf"
```

### How It Works

- PDF read hoti hai page by page
- Text ko 800-character chunks mein split kiya jaata hai
- Har chunk ka embedding banaya jaata hai
- ChromaDB mein store hota hai
- Query aane pe most relevant chunks retrieve hote hain
- Llama3 ko context ke roop mein diya jaata hai

### Books Added in This Build

- Atomic Habits — James Clear (662 chunks)
- The Psychology of Persuasion
- Thinking Fast and Slow — Daniel Kahneman
- The Castle — Franz Kafka
- The Trial — Franz Kafka
- Metamorphosis — Franz Kafka

# Chapter 11: Common Errors and Fixes

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## **Error: ModuleNotFoundError: torchaudio**

- Fix: voice.py mein koi advanced version paste hua. Clean voice.py replace karo.

## **Error: SyntaxError: return with value in async generator**

- Fix: brain.py ya memory.py corrupt hai. Clean version se replace karo.

## **Error: cannot import name 'PCControlModule'**

- Fix: pc\_control.py mein wrong version hai. Clean version se replace karo.

## **Error: ERR\_NGROK\_8012**

- Fix: ALTROS server chal nahi raha. python main.py pehle chalao, phir ngrok http 8000.

## **Error: Port 8000 already in use**

- Fix: Pichla ALTROS process band nahi hua. taskkill /f /im python.exe run karo.

## **Error: Knowledge base 0 chunks**

- Fix: Books dobara add karni hain — --add-book command use karo.

## **Error: Module skipped (disabled)**

- Fix: config.py mein MODULES\_ENABLED mein sab True karo.

## **Error: Ollama connection error**

- Fix: ollama serve command run nahi hua. Terminal mein chalao.

## **Error: Instagram login fail**

- Fix: instagram keyword PC Control ne pakad liya. pc\_control.py mein APPS dict se 'instagram' remove karo.

## **Error: Gmail: credentials.json not found**

- Fix: Google Cloud Console se download karke C:\ALTROS\ mein rakho.

## Chapter 12: Advanced Improvements (Future)

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### Short Term (1-2 weeks)

- Oracle Cloud free tier deploy — 24/7 online bina laptop ke
- GitHub repository setup — version control
- Google Calendar integration — schedule management
- YouTube module — trending videos, search
- Better voice — Edge TTS replace pyttsx3

### Medium Term (1-2 months)

- Auto-learning — conversation se automatically facts extract karo
- Multi-model support — coding ke liye CodeLlama, reasoning ke liye Mistral
- WhatsApp daily briefing — ALTROS khud se update bheje
- News aggregator — personalized daily news
- Spending tracker — expenses monitor karo

### Long Term (3-6 months)

- Proper Android app — Flutter se native mobile app
- Multi-account Gmail — sab accounts ek jagah
- Business intelligence — sales, expenses, profit tracking
- AI-powered content creation — Instagram captions, emails
- Voice clone — tumhari awaaz mein ALTROS bole
- Self-improving memory — important memories automatically prioritize

## Chapter 13: Complete Technology Stack

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| Technology       | Version    | Purpose             | Cost |
|------------------|------------|---------------------|------|
| Python           | 3.14       | Main language       | Free |
| Ollama           | Latest     | Local AI engine     | Free |
| Llama3           | 8B (4.7GB) | AI brain model      | Free |
| ChromaDB         | Latest     | Vector database     | Free |
| Whisper          | small      | Voice recognition   | Free |
| pyttsx3          | Latest     | Text to speech      | Free |
| pyporcupine      | Latest     | Wake word           | Free |
| DDGS             | Latest     | Web search          | Free |
| FastAPI          | Latest     | Web server          | Free |
| Uvicorn          | Latest     | ASGI server         | Free |
| ngrok            | Latest     | Tunnel/phone access | Free |
| pyautogui        | Latest     | PC control          | Free |
| Node.js          | v24        | WhatsApp runtime    | Free |
| whatsapp-web.js  | Latest     | WhatsApp API        | Free |
| instagrapi       | Latest     | Instagram API       | Free |
| Google Gmail API | v1         | Gmail integration   | Free |
| keyboard         | Latest     | Hotkeys             | Free |

**Total Cost: Rs. 0 (Completely Free)**

# ALTROS

Your Personal AI Brain — Complete and Running

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This system was built from scratch by Manish Kuntal, BTech CSE student, Mathura UP. It demonstrates that a world-class personal AI assistant can be built using entirely free, open-source tools with zero cloud dependency and complete privacy.

## Key Achievements

- 100% local — no cloud, no subscription, no privacy risk
- Full voice conversation with interrupt detection
- Real-time internet search with smart filtering
- Complete PC control — 35+ apps, files, system
- Phone access from anywhere via Jarvis UI
- WhatsApp, Gmail, Instagram — all integrated
- Book learning with 662+ knowledge chunks
- Auto-startup — boot laptop, ALTROS starts automatically